Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/25/2005
PI Name:	Loewenberg, Michael Ph.D.		
Project Title:	Flows of Wet Foams and Concentrated Emulsions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8286	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	07/01/2000	**End Date:**	07/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2477		
Performance Goal No.:			
Performance Goal Text:			

Through the development of detailed numerical simulations, this research is designed to complement ongoing efforts to explore the behavior in these systems experimentally and to develop theoretical descriptions. The long-range goal of the proposed research is the formulation of reliable microphysically-based statistical models. To progress effectively in this direction, the proposed investigation involves the following tasks and sub-tasks:
1. Concentrated emulsions in shear flow

- Develop computer simulations of concentrated emulsions in stationary shear.

- Compute the shear-thinning viscosity and normal stresses in flowing systems at high disperse-phase volume fractions. - Explore the influence of adsorbed surfactants on the rheology of dense emulsions.

Task Description:	- Determine critical conditions for drop breakup in concentrated emulsions. 2. Frequency response of wet foams and concentrated emulsions in oscillatory shear - Develop boundary integral simulations to describe the dynamics of systems with finite yield stresses. - Determine the linear viscoelastic response for systems with high dispersed-phase volume fractions. - Explore the nonlinear frequency response and microstructural evolution for higher strain. The aim of the proposed research is to advance a microstructural understanding of foam and emulsion flows. The focus is on the behavior of these systems in the intermediate regime of wet foams and dense emulsions about which the least is known. The proposed investigation focuses on the relation between the rheology of a foam or emulsion and its microstructure. This relation underlies important practical issues that lie in two general categories. 1. Inferring and manipulating the microstructure of foams and emulsions - Which microstructural features of a foam or emulsion be reliably inferred from rheological measurements? - Which microstructural parameters can be controlled by flow processing? - How can these objectives be achieved? 2. Predicting and controlling foam and emulsion rheology - How can the rheology of a foam (emulsion) be reliably predicted given some knowledge of its microstructure? - What microstructural features most affect the rheology? - To what extent can emulsified fluids with a particular rheology be designed? The proposed investigation will address these questions; therefore, the results should have direct, practical application in diverse fields such as materials processing, food processing, cosmetics, and pharmaceutical manufacturing. More generally, the research is designed to complement ongoing experimental and theoretical studies. The long-range goal is a fundamental understanding of the rheology and microstructural evolution in these systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Completing three-dimensional boundary integral simulations. Publishing theory for foam drainage in the presence of interfaces with Marangoni stresses.
Bibliography Type:	Description: (Last Updated: 10/07/2019)
Articles in Peer-reviewed Journals	Cristini V, Blawzdziewicz J, Loewenberg M, Collins LR. "Breakupin stochastic Stokes flows: sub-Kolmogorov drops in isotropic turbulence." J Fluid Mech. 2003 Oct 10;492:231-50. https://doi.org/10.1017/S0022112003005561 , Oct-2003
Articles in Peer-reviewed Journals	Cunha FR, Loewenberg M. "A study of emulsion expansion by a boundary integral method." Mech Res Commun. 2003 Nov-Dec;30:639–49. https://doi.org/10.1016/S0093-6413(03)00068-5 , Nov-2003
Articles in Peer-reviewed Journals	Nemer MB, Chen X, Papadopoulos DH, Blawzdziewicz J, Loewenberg M. "Hindered and enhanced coalescence of drops in stokes flows." Phys Rev Lett. 2004 Mar 19;92(11):114501. 2004 Mar 19;92(11):114501. https://doi.org/10.1103/PhysRevLett.92.114501 ; PubMed PMID: 15089142 , Mar-2004
Articles in Peer-reviewed Journals	Ismail AE, Loewenberg M. "Long-time evolution of a drop size distribution by coalescence in a linear flow." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Apr;69(4 Pt 2):046307. https://doi.org/10.1103/PhysRevE.69.046307 ; PubMed PMID: 15169099 , Apr-2004